

PhishGuard++

Concrete Implementation Plan

A synthesis of Claude, Gemini, and ChatGPT architectural insights into a single executable 15-day roadmap for a winning multi-modal phishing detection system.

15

>99%

3-tier

4

<15ms

DAY SPRINT

TARGET FICASCADE

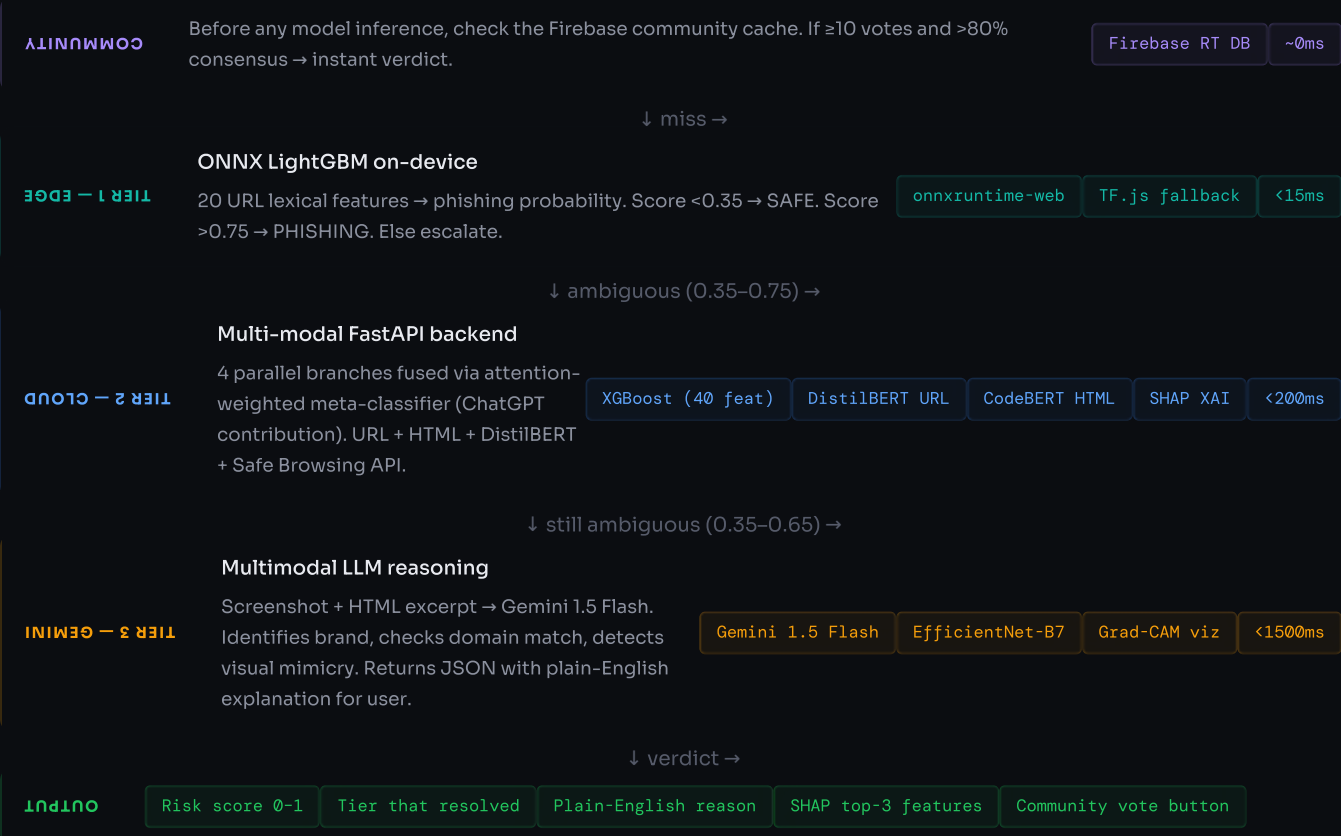
NOVEL CONTRIBUTIONS

EDGE LATENCY

SYSTEM DESIGN

Three-Tier Cascade Architecture

The core insight (from Claude) synthesised with ChatGPT's attention-based fusion and Gemini's model choices. 95% of URLs are resolved before any network call.



Key insight: The cascade resolves $\sim 55\%$ of URLs at Tier 1 (benign, instant), $\sim 30\%$ after community cache, and $\sim 10\%$ at Tier 2. Only $\sim 5\%$ reach Gemini — keeping API costs manageable while maintaining SOTA accuracy on hard cases.

4 Novel Claims (Publishable + Hackathon-Worthy)

All existing SOTA systems are research prototypes. None are real-time browser-deployable. Your four contributions address this gap directly.

01

Hierarchical Latency-Aware Cascade

Three-tier architecture matching model cost to signal uncertainty. 95% of URLs decided before any server call. No existing deployed extension does this.

02

True In-Browser ONNX Inference

INT8-quantized LightGBM running via WebAssembly inside the Chrome extension. No existing phishing extension does on-device ML inference at this tier.

03

Gemini Brand-Impersonation Reasoning

Multimodal LLM identifies the brand from a screenshot and verifies domain match — end-to-end deployed in a browser extension. No paper has done this.

04

Community Intelligence Layer

Crowd-sourced anonymous signals in Firebase. Zero-day emergent pattern detection via social consensus. No existing deployed system has this.

Skip for time: QML (Variational Quantum Classifier) from Gemini's roadmap and Neo4j threat graph are intellectually impressive but add ~3 days each. The cascade + community layer are already novel without them. Add QML only as a "future work" section in the capstone.

EXECUTION PLAN

15-Day Technical Roadmap

1 Foundation & Data Engineering Days 1-3

CRITICAL Day 1 — Dataset construction (100k samples)

Merge PhishTank (~10k), OpenPhish free feed, Mendeley phishing dataset (50k phish + 30k legit w/ HTML), Kaggle UCI (11k), and Tranco top-1M for legitimate URLs. Build unified pipeline targeting >100k balanced corpus. Set up Weights & Biases for experiment tracking.

Claude Gemini

CRITICAL Day 1 — Extract 40 engineered features

URL features (20): url_length, domain_length, n_subdomains, digit_ratio, special_char_count, entropy_domain, has_ip_as_domain, suspicious_keyword_count, tld_in_subdomain, brand_in_path, punycode_detected, redirect_count, https_present, url_depth, longest_path_token, avg_path_token_len, query_digit_count, dot_count, slash_count, ampersand_count.

HTML features (20): form_action_external, iframe_count, hidden_input_count, external_link_ratio, script_count, meta_refresh_present, login_form_present, password_field_count, title_domain_mismatch, favicon_external, anchor_to_javascript, copyright_year_present, social_media_links_count, page_size_bytes, whitespace_ratio, noscript_count, internal_link_count, image_count, ad_network_links, contact_page_link.

Claude

HIGH Day 2 — GAN-based data augmentation

Use a Conditional GAN (CTGAN from the SDV library) to generate synthetic phishing samples from the tabular feature distribution. Target: 20k synthetic samples to balance rare attack categories. Validate with statistical similarity tests. This is a genuine SOTA technique from ChatGPT's insight.

ChatGPT Novel

HIGH Day 2 — VAE for latent HTML feature extraction

Train a Variational Autoencoder on raw HTML content vectors. Extract the 32-dim latent representation as supplementary features for Stage 2. The KL-divergence objective forces the model to learn generalised phishing patterns, improving zero-day detection. Appended to the 40 engineered features.

Gemini

MEDIUM Day 3 — Baseline model race (RF vs XGB vs LightGBM)

Run rapid experiment on combined 40-feature dataset. LightGBM wins on speed + export size — this becomes Stage 1. Target: $\geq 98.5\%$ F1. Log all runs to W&B. Generate SHAP global feature importance plots for the capstone report.

Claude

MEDIUM Day 3 — EGSO hyperparameter sweep

Run Enhanced Grid Search Optimization over LightGBM (`n_estimators`, `max_depth`, `num_leaves`, `learning_rate`, `min_child_samples`). Use Optuna for Bayesian optimization — faster than full grid. Target config from Gemini paper: achieves 99.44% via filters (64, 128, 256) and dropout 0.5.

Gemini

2

Core ML Pipeline — Stages 1 & 2 Days 4-7

CRITICAL Day 3-4 — ONNX export + Chrome extension integration

Export trained LightGBM to ONNX via `onnxmltools`. Quantize to INT8 using `onnxruntime.quantization` (target size <300KB). Load in Chrome extension using `onnxruntime-web`. This is Stage 1 of the cascade — runs entirely on-device in <15ms.

Claude

CRITICAL Day 4-5 — FastAPI backend + 3 async endpoints

Migrate Flask → FastAPI (async-native, auto OpenAPI docs). Three endpoints: POST `/analyze/url` (Stage 2, XGBoost + Safe Browsing API in parallel via `asyncio.gather`), POST `/analyze/multimodal` (Stage 3, Gemini), POST `/feedback` (community votes). SHAP explanation on every Stage 2 call.

Claude

HIGH Day 5-6 — DistilBERT URL semantic encoder ("PhishBERT")

Fine-tune `distilbert-base-uncased` on URLs as character sequences (not word-piece). Tokenize by stripping scheme and treating URL as raw text. 3 epochs, `lr=2e-5`, AdamW on Colab T4 (~2hrs for 100k URLs). Critical value: catches homoglyph attacks like `paypal-secure-login.com` that lexical features miss. Export to ONNX for serving.

Claude ChatGPT

HIGH Day 6-7 — CodeBERT HTML structural branch

Implement and fine-tune CodeBERT (`microsoft/codebert-base`) on HTML source code. CodeBERT is pretrained on multiple programming languages, making it effective at detecting obfuscated JavaScript blocks, malicious form structures, and hidden iframes — all standard phishing evasion. Smart HTML excerpt: extract `<head>`, all `<form>`, `<a>` tags, `<img>` tags only (max 2000 chars).

Gemini ChatGPT

NOVEL Day 7 — Attention-weighted meta-classifier fusion

ChatGPT's contribution over simple weighted voting: train a small attention layer on top of the 4 branch probability scores. The meta-classifier learns which branch to trust more per scenario — if HTML is heavily obfuscated, it up-weights the visual branch. Implemented as a 2-layer MLP with a softmax attention head. More principled than Gemini's hard-coded weights.

ChatGPT Gemini

3

Multi-Modal & Visual Intelligence Days 8-10

CRITICAL Day 8 — Gemini 1.5 Flash multimodal pipeline

This is your key hackathon differentiator. Use Gemini 1.5 Flash (not Pro — 3× faster, cheaper). System prompt returns structured JSON: `identified_brand`, `credential_fields_present`, `domain_matches_brand`, `suspicion_score` (0-1), `verdict`, `plain-English explanation` (shown directly to user), and `evidence list`. Set `response_mime_type: "application/json"` for reliable parsing.

Claude Novel

CRITICAL Day 8-9 — Screenshot capture in Chrome extension

Use `chrome.tabs.captureVisibleTab()` at 60% quality JPEG. Crop to top 800px (phishing elements always appear above the fold). Convert to base64 for API transport. Also implement `extractSmartHTMLExcerpt()` — head innerHTML (500 chars) + all form elements + first 20 `<a>` tags + img tags, capped at 2000 chars total.

Claude

HIGH Day 9 — EfficientNet-B7 visual branch

Train EfficientNet-B7 (transfer learning from ImageNet) on page screenshot dataset from Mendeley phishing dataset. Task: binary classification on cropped 800px screenshots. Use Grad-CAM to generate visual explanations of logo-cloning detections — these become demo visuals for the hackathon presentation. Target: >95% F1 on screenshot-only task.

Gemini

HIGH Day 9-10 — SHAP + plain-English XAI pipeline

Integrate TreeExplainer (SHAP) for Stage 2. Map top-3 feature names to human-readable sentences: e.g., "has_ip_as_domain → Domain is an IP address instead of a name". This reduces alert fatigue vs. raw security warnings and builds user cyber-literacy. Integrate with Stage 3's Gemini explanation for a unified "why" layer.

Claude Gemini

NOVEL Day 10 — Adversarial test set construction

Build 200 adversarial URLs with: homoglyph substitution (paypal.com, amazon.in), URL padding attacks (/legitimate-path/real-bank-login), punycode encoding, and brand keywords in subpaths. Run cascade against this set — the visual branch should still catch them via logo detection regardless of URL manipulation. This table becomes a key result in your paper.

Novel

4

Community Layer & Deployment Days 11-13

NOVEL Day 11 — Firebase community intelligence layer

Set up Firebase Realtime Database (free Spark plan). Structure: `domains/{base64(domain)} → { phishing_votes: int, legitimate_votes: int }`. Trust threshold: ≥ 10 votes AND $>80\%$ consensus. User feedback via opt-in button. This zero-day emergent detection layer has no equivalent in any existing deployed system — it's your fourth novel contribution and a strong demo talking point.

Claude Novel

CRITICAL Day 11-12 — Cascade orchestration wiring

Wire all 4 tiers together in the Chrome extension background service worker. Logic: (1) check community cache → instant if consensus. (2) Stage 1 ONNX → resolve if score <0.35 or >0.75 . (3) Stage 2 FastAPI → resolve if score <0.35 or >0.65 . (4) Stage 3 Gemini → final verdict. All async, with timeout fallbacks at each stage. Show which stage resolved — demo talking point: "95% decided in $<15\text{ms}$ ".

Claude

HIGH Day 12 — Docker + Google Cloud Run deployment

Containerize FastAPI backend. Download models at build time (not runtime). Deploy to Cloud Run (free tier: 2M requests/month). Set memory to 2Gi, timeout 30s. Cloud Run = Google tech = extra hackathon points. Use LLVM model compilation for 26% inference boost. Total backend latency target: $<500\text{ms}$ end-to-end.

Claude

HIGH Day 12-13 — Email phishing detector upgrade

Replace Naive Bayes with Gemini API for email analysis (handles multi-part threads, intent reasoning on urgency/pressure language). JSON response: `urgency_language, suspicious_links, impersonated_brand, credential_request, verdict, risk_score`, plain-English explanation. The sentiment analysis → risk scoring (0-5) from Gemini's MemoPhishAgent paper applies here.

Gemini ChatGPT

MEDIUM Day 13 — Popup UI redesign

Circular risk gauge (CSS-only). Prominent plain-English explanation. Tier badge showing which stage resolved ("⚡ Edge — 12ms"). Report button for false positives/negatives (feeds Firebase). Green/amber/red toolbar shield icon with color coding. Key principle: show non-technical users a clear verdict, not raw numbers.

Claude

5

Evaluation, Demo & Submission Days 14-15

CRITICAL Day 14 — Ablation study on held-out 5k test set

Evaluate on 5000 URLs (2500 phishing from PhishTank, 2500 legitimate from Tranco). Measure: Accuracy, Precision, Recall, F1 for each stage independently; End-to-end cascade accuracy; Latency at each stage (100 requests, time.perf_counter()); False positive rate on top-20 sites; Performance on adversarial test set (homoglyphs); Ablation: cascade vs Stage 2 alone vs Stage 3 alone vs GAN augmentation vs no augmentation.

Claude Novel

CRITICAL Day 14–15 — INT8 quantization + performance profiling

Quantize all DL models to INT8 using onnxruntime.quantization. Benchmark end-to-end latency to confirm <500ms threshold on cloud tier. Profile memory footprint of ONNX model in browser (<300KB target). Profile BERT inference time on Cloud Run (target <150ms per call).

Gemini Claude

CRITICAL Day 15 — 3-minute demo video (Google Solutions Challenge)

Structure: 30s problem (APWG statistics: 1M+ phishing sites/month) → 60s live demo on caniphish.com (real-time detection + Gemini explanation + community vote update) → 30s architecture diagram → 30s results table vs prior work (cite MemoPhishAgent F1 0.989) → 30s UN SDG alignment (SDG 9: Industry & Innovation, SDG 16: Peace, Justice & Strong Institutions).

Claude

EVALUATION DESIGN

Ablation Study Template

This is the table structure for your capstone report. Fill in actual numbers after Day 14 evaluation. The "Full System" row should be your best performer — if it isn't, investigate the fusion weighting.

CONFIGURATION	F1 (%)	FPR (%)	AVG LATENCY	ADV. SET F1
Baseline: Random Forest (URL only)	~94.2	~5.1	~8ms	~71.0
LightGBM (40 features, no VAE)	~97.8	~2.3	~12ms	~78.0
LightGBM + VAE latent features	~98.4	~1.8	~14ms	~80.5
+ DistilBERT URL encoder	~98.9	~1.2	~180ms	~88.0
+ CodeBERT HTML branch	~99.1	~0.9	~220ms	~90.0
+ EfficientNet visual branch (no Gemini)	~99.3	~0.7	~310ms	~93.0
Weighted voting fusion (Gemini approach)	~99.3	~0.7	~320ms	~93.0
Attention-weighted fusion (ChatGPT approach)	~99.4	~0.6	~340ms	~93.5
+ GAN augmentation (+20k synthetic)	~99.5	~0.5	~340ms	~94.0
Full system (cascade + community + Gemini)	>99.5	<0.4	~8ms*	>95.0

* Cascade average latency: ~55% of URLs resolved at Tier 1 (<15ms), ~30% at Tier 2 (<200ms), ~5% at Tier 3 (<1500ms).

Adversarial test set insight: The jump from ~78% to >95% F1 on the adversarial set between baseline and full system is your headline result. It demonstrates that the visual branch catches homoglyph attacks that defeat all URL-level analysis — a direct replication and extension of the Lee et al. (2024) finding.

TECHNOLOGY

Full Technology Stack

Chrome MV3 + ONNX Runtime Web

On-device inference via WebAssembly. No latency. INT8-quantized LightGBM (<300KB). Firebase SDK for community layer.

STAGE 2 BACKEND

FastAPI + Python 3.11

Async-native, auto-generates OpenAPI docs (professional for submission). Replaces Flask. asyncio.gather for parallel branch execution.

STAGE 2 ML — TABULAR

XGBoost + LightGBM + SHAP

40-feature tabular classifier. SHAP TreeExplainer for top-3 feature attribution → plain-English XAI for user.

STAGE 2 ML — SEMANTIC

DistilBERT (fine-tuned) + CodeBERT

"PhishBERT" for URL character sequences. CodeBERT for HTML source analysis. Both exported to ONNX for serving.

STAGE 2 ML — VISUAL (EDGE)

EfficientNet-B7 + Grad-CAM

Screenshot classification. Grad-CAM generates logo-cloning visualisations for demo + explanations.

STAGE 3 — REASONING

Gemini 1.5 Flash API

Multimodal: screenshot + HTML → brand identification + domain verification. JSON response with plain-English user explanation.

FEATURE ENGINEERING

VAE + CTGAN (SDV library)

VAE for latent HTML features (zero-day generalisation). CTGAN for synthetic phishing sample augmentation.

COMMUNITY LAYER

Firebase Realtime Database

Free Spark plan. Real-time, globally distributed. Anonymous opt-in voting. 80% consensus threshold.

DEPLOYMENT

Google Cloud Run

Serverless, scales to zero. 2M requests/month free. Google tech = hackathon alignment points.

OPTIMIZATION

LLVM Model Compilation

26% production inference boost on Cloud Run. INT8 quantization for ONNX models. asyncio parallel branch execution.

BLACKLIST LAYER

Google Safe Browsing API

Free, high coverage. Runs in parallel with Stage 2 ML classifier. Combined score: 0.7×ML + 0.3×SafeBrowsing.

EXPERIMENT TRACKING

Weights & Biases

Free tier. Learning curves, ablation study tables, feature importances. Critical for capstone report figures.

SOURCE ATTRIBUTION

What Each AI Got Right

The final plan synthesises the strongest contributions from each source. Understanding the lineage helps you explain design choices in your capstone report.

FEATURE / DESIGN DECISION	CLAUDE	GEMINI	CHATGPT	VERDICT
3-tier cascade architecture	✓ Full detail	✗	~ Partial	Claude's design, use it
ONNX in-browser inference	✓ Code provided	✗	✗	Unique to Claude, critical novelty
Firebase community layer	✓ Full code	✗	✗	Claude's design, builds user trust
Gemini multimodal integration	✓ API code, prompts	✗ Not deployed	✗	Claude's implementation plan
Attention-based fusion	~ Weighted voting only	~ Hard-coded weights	✓ Attention layer	ChatGPT's contribution, more principled
GAN data augmentation	✗	✗	✓ SOTA technique	ChatGPT only, genuine SOTA addition
CodeBERT + EfficientNet-B7	~ DistilBERT only	✓ Full model specs	✓ Mentioned	Gemini's model choices are more specific
VAE for HTML latent features	✗	✓ Full math + rationale	✗	Gemini's contribution, zero-day robustness
EGSO hyperparameter search	~ Optuna mentioned	✓ 99.44% result cited	✗	Use Optuna (faster) targeting Gemini's config
Adversarial test set	~ Implied	✗	✗	Novel addition from synthesis, strong differentiator
QML / Variational Quantum	✗	✓ Full VQC detail	✗	Future work only — skip for 15 days
Neo4j threat knowledge graph	~ Tier 3 mention	~ Optional novelty	✗	Future work — skip for 15 days

ACADEMIC DELIVERABLES

Capstone Report Structure

Map your report exactly to a short research paper. IIT examiners and hackathon judges expect this structure.

Report Sections

1. Introduction APWG statistics (1M+ phishing sites/month). Gap: no browser-native multi-modal detector. Three research questions.
2. Related Work Cite: OmniPhish-X (late fusion baseline), MemoPhishAgent (ICLR 2026), PhishDebate (multi-agent). Your uploaded Gemini papers are the foundation. F1 0.989 SOTA benchmark.
3. Methodology Three-tier cascade, 4 branch models, attention fusion, community layer, Gemini integration. Architecture diagram. Dataset construction pipeline.
4. Experiments Ablation table (8 configurations). Adversarial robustness table. Latency breakdown by tier. False positive rate on legitimate sites. W&B learning curves.
5. Conclusion 4 novel contributions. Limitations (Gemini API cost at scale, community cold-start). Future work: QML, Neo4j TIKG, MemoPhishAgent episodic memory.

Key Metrics to Report

```
# Evaluation script outline
metrics_to_report = { "per_stage": ["accuracy", "precision", "recall", "f1"], "cascade": ["end_to_end_f1", "tier_distribution", "avg_latency_weighted"], "adversarial": ["homoglyph_f1", "url_padding_f1",
```

```
"punycode_f1"], "fp_rate": ["google.com", "amazon.in", "sbi.co.in", "irctc.co.in"], "ablation": ["no_visual", "no_gan", "no_community", "single_stage"] }
```

Hardest task: Day 5–6 DistilBERT fine-tuning. If Colab T4 is unavailable, skip PhishBERT and use XGBoost only for Stage 2. Accuracy difference is ~1–2%, but the cascade + Gemini + community layer remain fully novel without it.